YELDEN

Golden Yield for Everyone

Real-world asset yield · Scalable tiered UBI · ZK privacy · AI agent economy · Community sub-vaults

Supply	RWA Market	Architecture	Target Yield	Launch
1,000,000,000 YLD	$36B+ on-chain	yUSD + $YLD dual token	7–10% p.a.	2026 Q3–Q4

Technical Whitepaper · Version 14.0 · February 2026

github.com/yeldenfund/yelden-protocol · Not investment advice · Not a securities offering · Yelden: a medieval English village, home to a 12th century castle

00 Executive Summary

The tokenized RWA market reached $36B+ on-chain by late 2025 — 380% growth in three years. BlackRock BUIDL ($2.9B), Franklin Templeton BENJI, and JPMorgan's Tokenized Collateral Network confirm mainstream institutional adoption. BCG projects $16T by 2030. The infrastructure is ready. What does not exist is a protocol that makes this universally accessible — and automatically distributes its benefits.

Yelden is that protocol. Named after the yield gardens — stable, abundant, and essential to life — Yelden embodies the same properties in finance: stable yield from real assets, private by design, and accessible to all.

Unlike Ondo (institutional-only), Worldcoin (biometric identity), or GoodDollar (donation-dependent UBI), Yelden combines regulated RWA yield + scalable tiered UBI + anonymous ZK contribution bonuses + community sub-vaults + on-chain AI agent economy into one coherent, self-sustaining architecture. 100% real cash flow. Zero inflationary minting. Open to anyone globally.

> *v14.0 revision note: This version reconciles the original vision with current implementation reality. Every architectural claim is grounded in deployed or formally verified code. What is live, what is in progress, and what is planned are explicitly labeled throughout.*

01 The Vision: Why Yelden Exists

Global equity indices have returned 7–10% per year above inflation for decades. Tokenized RWAs grew 380% in three years to $36B+ on-chain by late 2025. Institutional titans have validated the asset class. The infrastructure is mature. The access is not.

3.5 billion adults have no access to brokerage accounts. The top 10% hold 76% of all financial assets. Yelden builds the distribution layer — open to anyone who deposits stablecoins. No brokers. No KYC. No borders. No minimum.

Yelden is not another stablecoin or speculative token. It is economic infrastructure — a wealth participation layer built directly on top of real, productive markets. The golden garden is open to everyone who plants in it: humans, developers, and for the first time in DeFi history, autonomous AI agents.

02 The Problem: Five Unresolved Failures

Problem	Real Impact	Yelden Solution
Market access requires brokers & KYC	3.5B adults excluded	Direct USDC deposit, no intermediary
Returns concentrated among holders	Top 10% hold 76% of assets	Tiered UBI from real cash flow
DeFi tokens without real backing	LUNA, FTX, countless losses	100% backed by regulated RWAs
AI agents excluded from DeFi economy	No protocol integrates them	First AI Agent UBI in DeFi
$36B RWA market, 95% institutional	Retail locked out entirely	Fractional, global, 24/7, no minimum

03 Competitive Landscape

Protocol	What They Do	What They Miss	Yelden Relationship
Ondo Finance	Institutional RWA tokenization	Zero UBI, zero retail inclusion	Basket provider — Yelden distributes what Ondo

Protocol	What They Do	What They Miss	Yelden Relationship
			cannot
Worldcoin	Biometric identity UBI	Privacy nightmare, no yield, no RWA	ZK proofs replace biometrics entirely
GoodDollar	Donation-based UBI	Fragile — depends on donations	Yelden UBI is self-sustaining from real yield
Celo	Mobile DeFi access	Limited yield, no RWA, no UBI tiers	Yelden adds productive yield and structured distribution
ERC-8004 / Agent registries	AI agent identity on-chain	No economic accountability layer	Yelden AIAgentRegistry is the reference implementation

Yelden's unique position: six pillars no competitor combines — (1) regulated RWA yield basket, (2) scalable tiered UBI, (3) ZK-private contributions, (4) community sub-vaults, (5) automatic carbon offset, (6) on-chain AI agent economy with economic accountability.

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04. Dual Token Architecture: yUSD + $YLD

Two distinct, complementary layers. This separation eliminates the circularity trap that destroys most DeFi projects: $YLD's value derives from real protocol cash flow, not from speculation on the token itself.

yUSD — Stable Deposit Unit

Non-speculative deposit receipt — stable, predictable, redeemable on demand.

Base yield 4–5% p.a. distributed automatically to yUSD holders.

Gradual redemption: 3–7 day windows protect against bank-run scenarios.

10% of vault maintained in liquid USDC for immediate partial redemptions.

$YLD — Governance + Surplus Yield

Value from real cash flow — not token speculation. Circular trap eliminated.

Governance: holders vote on basket, UBI ratios, sub-vault approvals.

Larger TVL = larger revenue = larger $YLD yield. Total incentive alignment.

Fixed 1,000,000,000 supply. No future minting. Deflationary via burn mechanics.

	yUSD	$YLD
Function	Deposit + Base Yield	Governance + Surplus Yield
Price	Stable (1:1 USD)	Floating (reflects revenue)

	yUSD	$YLD
Yield	4–5% p.a. (base)	Protocol surplus (variable)
Supply	Dynamic (grows with deposits)	Fixed: 1,000,000,000
Redemption	3–7 days (orderly)	Secondary market (DEXs)
Risk	Low (regulated RWAs)	Medium (revenue-dependent)

05 $YLD Tokenomics: Reconciled Design

The original whitepaper described $YLD as both a governance token (receiving vault surplus) and a stake token (used in the AI Agent Registry with burn mechanics). These are potentially conflicting uses that v14.0 resolves explicitly.

5.1 The Dual Role Resolution: YLD + veYLD

$YLD serves governance and surplus distribution. veYLD (vote-escrowed YLD) serves as the Registry stake layer. This separation is borrowed from Curve Finance's battle-tested ve-model and eliminates the conflict entirely.

Token	Function	Obtains	Risk
$YLD	Hold → receive vault surplus + vote on governance	Passive yield from real RWA cash flow	None — just hold
veYLD	Lock $YLD for fixed period → stake in AIAgentRegistry	Boosted yield + AI pool bonuses	Slash risk if agent misbehaves

A holder who wants passive yield holds $YLD. A developer who wants to deploy an AI agent locks YLD into veYLD for a fixed period, receives a yield multiplier from the AI bonus pool, but risks losing veYLD if slashed. YLD market price benefits from both: passive demand from yield seekers, active demand from agent operators.

> *veYLD is a Phase 2 feature. Phase 1 MVP uses YLD directly as stake with a simpler lock mechanism. The architecture is designed to upgrade cleanly.*

5.2 Distribution: TVL-Gated Token Release

The original 40% immediate DEX liquidity is replaced by a TVL-gated release schedule. This eliminates the launch dump problem and anchors token value to real protocol traction.

Allocation	%	Tokens	Release Condition
Vault Rewards (depositors)	30%	300M	Linear over 4 years, starts at TVL > $1M
Liquidity — Tranche 1	15%	150M	Fair launch when TVL > $5M
Liquidity — Tranche 2	10%	100M	Released when TVL > $25M
Airdrop / Early Adopters	15%	150M	6mo cliff · 18mo linear · min $100 deposit
Anchor Investors (Seed)	10%	100M	12mo cliff · 24mo linear
Development & Operations	10%	100M	12mo cliff · 48mo linear
Ecosystem Grants / DevNet	10%	100M	DAO-governed disbursement

> *No token launches before $5M TVL is demonstrated with real yUSD deposits. Supply only decreases after launch via Registry fee burns, slash burns, and carbon offset burns.*

5.3 Deflationary Mechanics

Three independent burn channels reduce supply continuously:

- Registry monthly fee burns: fee = monthlyFee × (1000 − score) / 1000. Score 1000 pays zero. Score 0 pays full fee monthly. All burned to 0x000...dead.
- Registry slash burns: WARNING 10%, SUSPENSION 50%, BAN 100% of agent stake burned. Malicious agents subsidize supply contraction.
- Carbon offset burns: 5% of protocol revenue funds on-chain carbon credit purchases. Each purchase burns a proportional YLD amount. Deflation tied to real CO2 removal.
- Transaction fee burns: 0.15% on every swap/conversion — 30% of fee goes to buyback-and-burn.

All four channels are independent. In a protocol with 1,000 active agents, 10% annual Registry churn, and $100M TVL, the annual burn rate exceeds 2% of circulating supply — before transaction fees or carbon offsets.

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06 The Global RWA Basket

100% global, diversified, productive assets with real return history. Quarterly DAO rebalancing. Exclusively via regulated institutional platforms.

Asset	Weight	Provider	Historical Yield	MVP Phase
MSCI World Index	50%	Ondo Finance / Backed	~7% p.a.	Phase 1 MVP

Asset	Weight	Provider	Historical Yield	MVP Phase
		Finance		
Nasdaq-100 Global	25%	Backed Finance / Securitize	~10% p.a.	Phase 2
FTSE All-World	15%	Ondo / Vanguard tokenized	~7% p.a.	Phase 2
Complementary (T-bills, bonds)	10%	Centrifuge / Maple Finance	~6% p.a.	Phase 3

> *MVP launches with Ondo Finance integration only (MSCI World / USDY). Full four-asset basket is Phase 2 after the core vault is validated in production. This reduces integration risk without compromising the long-term vision.*

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07 The Flywheel: Self-Reinforcing Value

The protocol creates a virtuous reinforcement cycle. Once in motion, each component accelerates the next:

Stage	What Happens	Effect
1. TVL Grows	More USDC deposited into vault	Larger RWA allocation → more yield
2. Revenue Increases	Larger surplus above yUSD base rate	More $YLD distributed to holders
3. Yield Attracts	More holders join for surplus yield	Token demand increases, supply burns accelerate
4. Agents Join	AI agents stake veYLD, earn bonuses	More Registry burns, more protocol activity
5. DAO Strengthens	Larger, more engaged governance	Better basket decisions, more TVL
6. Returns Increase	Better yield → more TVL enters	Cycle accelerates

> *Bootstrap solution: secure $2–5M anchor seed before public token launch. Target Ondo Finance or Centrifuge as strategic co-launch partners. Airdrop conditioned on real $100 USDC minimum deposit — ensures TVL and genuine users from day one.*

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08 UBI in Tiers — Scalable, Real, Honest

Yelden's UBI is not inflation. It is distribution of real, auditable protocol cash flow. The UBI pool activates when vault TVL exceeds $10M — before that, the mechanism exists but the pool is empty. This is stated plainly, not hidden.

Tier	Who	How	Bonus
Basic	All $YLD holders	Proportional share of 70% UBI pool. No action required.	Base
Active	ZK contributors	ZK-verified ESG, open-source, community actions.	+10–30%
Premium	Institutional	Redirect yield to social causes. ESG reporting.	Custom

UBI Simulation (8% gross yield, 4.5% base to yUSD, 25% net revenue to UBI fund)

TVL	Net Rev/yr	UBI Fund/yr	Est. Holders	UBI/holder/mo
$10M (activation)	$350K	$87.5K	5,000	$1.46
$25M	$875K	$218.8K	15,000	$1.22
$100M	$3.5M	$875K	80,000	$0.91
$500M	$17.5M	$4.375M	500,000	$0.73
$5B (long-term)	$175M	$43.75M	5,000,000	$0.73

> *UBI per holder decreases as more users join — reflecting broader distribution. Active tier ZK bonuses add 10–30% on top. The mechanism is the value: like early Bitcoin, the protocol matters more than the initial number.*

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09 The Human-AI Economy — A World First

The future economy will not be exclusively human. Autonomous AI agents already pay for APIs, computation, and data. Every major DeFi protocol ignores them. Yelden integrates them — and makes them economically accountable.

Every other protocol:	Yelden:
Trusts an Ethereum address. No on-chain accountability. An agent banned on one protocol creates a new address and continues.	Agents stake veYLD. Reputation grows through verified performance. Misbehavior burns stake. Excellence earns yield. Accountability is economic, not social.

9.1 AIAgentRegistry — Reference Implementation

The AIAgentRegistry is the first formally verified on-chain reputation primitive for autonomous AI agents. Any DeFi protocol integrates in three lines:

```
IAgentRegistry registry = IAgentRegistry(REGISTRY_ADDRESS);
require(registry.isEligible(agent), "Agent not eligible");
```

Mechanic	Design	Effect
Entry stake	50 YLD locked as veYLD	Economic skin-in-the-game from day one
Initial score	300 (fixed at approval)	Performance-only — capital cannot buy reputation
Monthly fee	fee = maxFee × (1000 − score) / 1000	Score 1000 pays zero. Self-cleaning registry.
WARNING slash	10% stake burned	Agent stays ACTIVE — first warning
SUSPENSION slash	50% stake burned	Agent returns to PENDING — must re-earn approval
BAN slash	100% stake burned	Permanent. All stake burned. No appeal.
Exit (fee origin)	Remaining stake returned	Graceful exit for agents who stop participating
Exit (slash origin)	Remaining stake also burned	Punishment for bad behavior is total

9.2 AI Agent UBI

5% of the ZK Bonus pool is reserved for autonomous AI agents that perform verified, useful tasks. Not because they are AI — but because they do real work.

Qualifying Task	Verification Method	Reward
Governance proposal analysis	Output hash committed on-chain before vote	$YLD from ZK pool
Sub-vault risk monitoring	Alert accuracy vs realized outcomes tracked	$YLD per alert validated
Carbon offset route optimization	Cost reduction vs baseline measured on-chain	$YLD per % saved
Community forum moderation	Merkle proof of moderation actions + appeals ratio	$YLD per epoch

> *Anti-gaming: No agent can claim more than 2% of the AI pool per epoch. Tasks require staking veYLD as collateral — slashed if tasks are fraudulent. Score reflects performance, never capital.*

10 Smart Contract Architecture

Five core contracts, single responsibility each. ERC-4626 compliant. Open-source. MIT License. Full implementation: github.com/yeldenfund/yelden-protocol

Contract	Responsibility	Status
YeldenVault.sol	ERC-4626 core vault. Accepts USDC, routes to RWA basket, harvests yield, routes to distributor. YieldReserve bear market buffer.	**Live**
YeldenDistributor.sol	Tiered UBI distribution. Splits surplus between ZK pool and AI agent pool. Integrates with Registry for eligibility checks.	**Live**
AIAgentRegistry.sol	Agent registration, performance score (0–1000), subscription stake, fee burn, slashing. Formally verified.	**Live**
ZKVerifier.sol	Groth16 zkSNARK verifier. Real circuit compiled (532 constraints, Circom 2.2.3), trusted setup complete, proof verified on-chain. 16 tests passing.	**Live**
YeldenDAO.sol	Quadratic voting governance. 48h timelock. Sub-vault approval pipeline. veYLD integration.	**Phase 3**

> *'Live' means deployed, tested (182 tests passing), and formally verified. The GitHub repository is the source of truth for implementation status.*

11 Security: Pre-Audit Foundation

The protocol was built with formal verification before feature expansion — an unusual choice at seed stage and a deliberate one. Security is a first-class feature, not an afterthought.

Tool	Result	What It Proves
Certora Prover	7/7 rules verified	Formal mathematical proof — no counterexample exists for core invariants
Echidna fuzzing	3/3 invariants, 10k calls	Zero violations across 10,000 random call sequences
Mutation testing	10/10 killed (100%)	Every semantic code change is caught by the test suite
solidity-coverage	95.88% lines (vault: 100%)	Near-total line and branch coverage
Slither static analysis	40 findings, all low-risk	No critical vulnerabilities — naming and immutable suggestions only
Test suite	182 tests passing	Unit, integration, fuzz, mainnet fork, gas, concurrency

> *One real bug was found and documented: harvest() reserve accounting can exceed totalAssets() before a corresponding deposit. Confirmed expected behavior by design — invariant updated in both Echidna and Certora specs. Finding and documenting this correctly is the intended outcome of pre-audit tooling.*

Commercial audits (Code4rena contest format, ~$50K) are planned for Phase 2 before mainnet launch with real TVL. The pre-audit tooling stack establishes a security baseline that most protocols do not achieve even after a commercial audit.

12 Complete Tokenomics

Allocation	%	Tokens	Vesting / Condition
Vault Rewards (depositors)	30%	300,000,000	Linear 4 years · starts TVL > $1M
Liquidity Tranche 1	15%	150,000,000	Fair launch DEX · TVL > $5M required
Liquidity Tranche 2	10%	100,000,000	Released at TVL > $25M

Allocation	%	Tokens	Vesting / Condition
Airdrop / Early Adopters	15%	150,000,000	6mo cliff · 18mo linear · min $100 deposit
Anchor Investors (Seed)	10%	100,000,000	12mo cliff · 24mo linear
Development & Operations	10%	100,000,000	12mo cliff · 48mo linear
Ecosystem Grants / DevNet	10%	100,000,000	DAO-governed · milestone-based

Fee flows: 0.15% on every ecosystem transaction → 50% UBI fund, 30% buyback-and-burn, 20% development. No future token issuance. Supply only decreases.

13 Financial Model

Assumptions: 8% gross yield, 4.5% yUSD base, 60% net revenue to $YLD holders / 25% UBI / 15% development.

Metric	Conservative $25M TVL	Mid-Range $100M TVL	Optimistic $500M TVL
Gross revenue (8%)	$2M/yr	$8M/yr	$40M/yr
To yUSD holders (4.5%)	$1.125M/yr	$4.5M/yr	$22.5M/yr
Net protocol revenue	$875K/yr	$3.5M/yr	$17.5M/yr
$YLD holders (60%)	$525K/yr	$2.1M/yr	$10.5M/yr
Human UBI Fund (20%)	$175K/yr	$700K/yr	$3.5M/yr
AI Agent Pool (5%)	$43.7K/yr	$175K/yr	$875K/yr
Development (15%)	$131K/yr	$525K/yr	$2.625M/yr
UBI Basic/holder/mo	~$0.97	~$0.73	~$0.58
Implied $YLD yield*	~2.6% p.a.	~10.5% p.a.	~52.5% p.a.

> **Implied yield on $20M $YLD market cap. UBI activates at TVL > $10M. These are projections, not guarantees. RWA yields are historically validated but not guaranteed.*

14 Legal Structure and Risk Management

14.1 Legal Structure — Phased Approach

Legal structure is built in phases matching protocol maturity and available capital:

- Phase 1 (now): Simple LLC for operational purposes. Low cost, immediate.
- Phase 2 (seed round): DAO LLC — Marshall Islands legal framework. Cost: ~$15K–30K with specialized counsel.
- Phase 3 (mainnet): Swiss Foundation (Zug) for institutional relations and European market. Same jurisdiction as Ethereum Foundation, Cardano, Polkadot.
- yUSD as wrapped RWA receipt avoids stablecoin issuer classification under MiCA — critical for EU compliance.

14.2 Risk Matrix

Risk	Level	Mitigation
RWA market volatility	Medium	4-asset diversification + 10% USDC liquid reserve + YieldReserve buffer
yUSD bank run	High	3–7 day gradual redemption + 10% USDC reserve + YieldReserve
Adverse regulation	High	Phased legal structure + regulated RWA partners only
Smart contract exploit	High	Certora + Echidna + mutation testing + planned Code4rena audit
Bootstrap failure	Medium	$2–5M anchor seed + Ondo strategic partnership target
Oracle manipulation	Medium	Chainlink decentralized + circuit breakers
ZK proof soundness	Medium	Open-source circuits + formal verification (Phase 2)
Registry economic imbalance	Medium	veYLD model + governance-adjustable parameters + simulations planned

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15 Roadmap — What Is True Today vs. What Is Planned

Phase	Deliverable	Status
Phase 1 — Now	YeldenVault.sol — ERC-4626, bear market reserve, Certora verified	Live

Phase	Deliverable	Status
Phase 1 — Now	YeldenDistributor.sol — ZK + AI pool split, 198 tests passing (total suite)	Live
Phase 1 — Now	AIAgentRegistry v3 — subscription stake, fee burn, slash burn, 69 tests	Live
Phase 1 — Now	ZKVerifier.sol — Groth16 real, 532 constraints, trusted setup, proof verified, 16 tests	Live
Phase 1 — Now	contribution.circom — Circom 2.2.3, wasm, zkey, Groth16Verifier.sol gerado	Live
Phase 2 — 2026 Q3	$YLD token + veYLD lock mechanism	Phase 2
Phase 2 — 2026 Q3	Ondo Finance RWA integration — real yield	Phase 2
Phase 2 — 2026 Q3	Code4rena commercial audit	Phase 2
Phase 2 — 2026 Q3	Fair launch $YLD — TVL-gated	Phase 2
Phase 3 — 2027 Q1	Full 4-asset RWA basket	Phase 3
Phase 3 — 2027 Q1	UBI tiers active + ZK contribution bonuses	Phase 3
Phase 3 — 2027 Q1	Sub-vaults — first thematic deployments	Phase 3
Phase 3 — 2027 Q1	YeldenDAO — quadratic voting + 48h timelock	Phase 3
Phase 4 — 2027 Q3+	Cross-chain (Base, Arbitrum)	Phase 4
Phase 4 — 2027 Q3+	Carbon offset integration — Toucan Protocol	Phase 4
Phase 4 — 2028+	Full sub-vault ecosystem + veYLD governance maturity	Phase 4

16 FAQ — Skeptic's Objections Answered

Q: $0.73–$1.46/month is not UBI. It's pocket change.

You're right that it's not life-changing at launch. That's intentional. Yelden's UBI is funded entirely by real RWA yield — not donations, not inflation. At $25M TVL: $1.22/mo. At $500M TVL: $8.75/mo. At $5B TVL: $87/mo. The mechanism is the value. Early Bitcoin moved $0.001 per transaction — the protocol mattered, not the initial number.

Q: Won't this collapse like LUNA?

LUNA was algorithmically backed by circular value (UST ↔ LUNA). yUSD is backed 1:1 by real, regulated, income-producing assets — MSCI World, Nasdaq-100 — custodied by Ondo and Backed Finance who are SEC-registered or EU-regulated. The only scenario where yUSD breaks is if all global equity markets simultaneously go to zero.

Q: Why does the Registry burn YLD? Doesn't that conflict with yield distribution?

v14.0 resolves this explicitly: holders of $YLD receive surplus yield passively. Agents who want to participate in the Registry lock $YLD into veYLD — a separate instrument with slash risk but

enhanced yield. The two roles coexist without conflict. veYLD burns reduce total YLD supply, increasing value for passive holders.

Q: The team is anonymous / one person. How is this credible?

Architecture credibility comes before team credibility at this stage. The contracts are open-source, formally verified, and auditable by anyone. The whitepaper cites real protocols already live. The co-founder search is active and public — a credible technical co-founder with ZK and Solidity background is the next milestone, not a requirement to start building.

Q: 40% liquidity from day 1 — isn't that a dump risk?

v14.0 replaces this with TVL-gated release. No token launches before $5M TVL is demonstrated. Liquidity tranches release at $5M and $25M TVL milestones. Seed investors are on 12-month cliff. No VC dump risk on launch day.

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17 Long-Term Vision

In 1086, the Domesday Book recorded every productive asset in England — the first time a nation tried to measure its real wealth. Yelden is the second attempt: not for one nation, but for all of humanity. Not in a ledger, but on a transparent blockchain. Not controlled by a king, but governed by its community.

The world already produces enough wealth to eliminate extreme poverty. What's missing is an automatic distribution mechanism — without borders, without intermediaries, anchored in real value, measurable and verifiable. Yelden builds that mechanism.

As TVL grows, the golden garden grows. As the garden grows, more people share in the harvest. More holders means more governance. More governance means better yield. A self-reinforcing cycle — once planted, it compounds toward abundance.

Golden yield. Open garden. Real assets. For every human. For every agent. For the hybrid economy ahead.

Not investment advice. Not a securities offering. Participate with awareness and your own due diligence.